

FAULT CODE 381 - Intake Air Heater

Number 1 Circuit Error

Warnings and Cautions

CAUTION

To reduce the possibility of damaging a new engine control module (ECM), all other active fault codes must be investigated prior to replacing the ECM.

CAUTION

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement: Part Number 3822758 - male Deutsch™/AMP™/Metri-Pack™ test lead and Part Number 3822917 - female Deutsch™/AMP™/Metri-Pack™ test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the fault codes.	
	STEP 1A. Check for an inactive fault code.	Fault Code 381 inactive?
STEP 2.	Check the intake air heater relay and circuit.	
	STEP 2A. Inspect the intake air heater relay and electrical connections.	Dirty or damaged pins?
	STEP 2B. Check for an open circuit in the intake air heater relay.	Less than 10 ohms?
	STEP 2C. Check the intake air heater diagnostic voltage, SUPPLY wire, and return circuit.	Greater than 3.75-VDC?
	STEP 2D. Check for an open circuit in the intake air heater relay return circuit.	Less than 10 ohms?

STEPS		SPECIFICATIONS
STEP 3.	Check the ECM and original equipment manufacturer (OEM) harness.	
	STEP 3A. Inspect the ECM and OEM harness connector pins.	Dirty or damaged pins?
	STEP 3B. Check for an open circuit in the OEM harness.	Less than 10 ohms?
	STEP 3C. Check for a pin-to-pin short circuit in the OEM harness.	Greater than 100k ohms?
	STEP 3D. Check for an inactive fault code.	Fault Code 381 inactive?
STEP 4.	Clear the fault code.	
	STEP 4A. Disable the fault code.	Fault Code 381 inactive?
	STEP 4B. Clear the inactive fault codes.	All fault codes cleared?

STEP 1. Check the fault codes.

STEP 1A. Check for an inactive fault code.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check for an inactive fault code. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 381 inactive? YES	Refer to Procedure 019-362 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html)
	Fault Code 381 inactive? NO	2A

STEP 2. Check the intake air heater relay and circuit.

STEP 2A. Inspect the intake air heater relay and electrical connections.

- Conditions :
- Turn keyswitch OFF.
 - Disconnect the intake air heater relays from the OEM harness.

Action	Specification/Repair	Next Step
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Inspect the OEM harness and intake air heater relay connections for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Perform these checks on all interconnecting harness connectors in the circuit. Refer to the circuit art or wiring diagram to identify the connectors and pins. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the sensor or harness connector. Repair the damaged harness, connector, or pins, if possible. • Flush the dirt, debris, or moisture from the connector pins. Use electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. • Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)	4A
	Dirty or damaged pins? NO	2B

STEP 2B. Check for an open circuit in the intake air heater relay.

Conditions :

- Turn keyswitch OFF.
- Disconnect the intake air heater relay from the OEM harness.

Action	Specification/Repair	Next Step
Check the intake air heater relay resistance. Measure the resistance between the intake air heater SIGNAL pin and the intake air heater RETURN pin at the intake air heater relay. Note: For 12 volt systems, both relays must be checked individually. Refer to the circuit diagram or the wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Less than 10 ohms? YES	2C
	Less than 10 ohms? NO Repair : An open circuit has been detected in the intake air heater relay. Replace the intake air heater relay. Refer to the OEM service manual.	4A

STEP 2C. Check the intake air heater diagnostic voltage, SUPPLY wire, and return circuit.

Conditions :

- Turn keyswitch OFF.
- Disconnect the intake air heater relay from the OEM harness.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
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Check the intake air heater supply voltage and return circuit. Measure the voltage between the intake air heater relay SIGNAL wire and the intake air heater relay RETURN wire at the intake air heater relay connector of the OEM harness. Note: For 12 volt systems, both leads must be tested.	Greater than 3.75-VDC? YES	3C
	Greater than 3.75-VDC? NO	2D

STEP 2D. Check for an open circuit in the intake air heater relay return circuit.		
Conditions : - Turn keyswitch OFF. - Disconnect the intake air heater relay from the OEM harness.		
Action	Specification/Repair	Next Step

	Less than 10 ohms? YES	3A
	Less than 10 ohms? NO Repair : An open circuit on the RETURN wire has been detected. Refer to the OEM wiring diagram for RETURN wire configuration. - If the RETURN is wired to the ECM, repair or replace the OEM harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html) If the RETURN wire is grounded to chassis or engine block ground, repair the source of the damaged connection. Clean, repair, or replace the OEM harness, if possible. - Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)	4A

Check for an open circuit.

- Measure the resistance between the intake air heater relay RETURN pin at the intake air heater relay connector to ground.

Refer to the circuit diagram or the wiring diagram for connector pin identification.

Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19.
(/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)

STEP 3. Check the ECM and OEM harness.

STEP 3A. Inspect the ECM and OEM harness connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the OEM harness from the ECM connector.

Action	Specification/Repair	Next Step
Inspect the OEM harness ECM connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Perform these checks on all interconnecting harness connectors in the circuit. Use the circuit art or wiring diagram to identify the connectors and pins. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the ECM or OEM harness connector. Repair the damaged harness, connector, or pins, if possible. • Flush the dirt, debris, or moisture from the connector pins. Use electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. • Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)	4A
	Dirty or damaged pins? NO	3B

STEP 3B. Check for an open circuit in the OEM harness.

Conditions :

- Turn keyswitch OFF.
- Disconnect the OEM harness from the ECM connector.
- Disconnect the intake air heater relay from the OEM harness.

Action	Specification/Repair	Next Step
Check for an open circuit. • Measure the resistance between the OEM harness ECM connector intake air heater relay SIGNAL pin and the OEM harness intake air heater relay connector SIGNAL wire. Note: For 12 volt systems, both leads must be tested. Refer to the circuit diagram or the wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Less than 10 ohms? YES	3C
	Less than 10 ohms? NO Repair : An open intake air heater relay signal circuit has been detected in the OEM harness. • Repair or replace the OEM harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)	4A

STEP 3C. Check for a pin-to-pin short circuit in the OEM harness.**Conditions :**

- Turn keyswitch OFF.
- Disconnect the OEM harness from the ECM connector.
- Disconnect the intake air heater from the OEM harness.

Action	Specification/Repair	Next Step
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Check for a pin-to-pin short. • Measure the resistance between the intake air heater relay SIGNAL pin in the OEM harness ECM connector and all other pins in the ECM connector. Refer to the circuit diagram or the wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? YES	3D
	Greater than 100k ohms? NO Repair : A pin-to-pin short circuit on the intake air heater relay SIGNAL wire has been detected in the OEM harness. • Repair or replace the OEM harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)	4A

STEP 3D. Check for an inactive fault code.

Conditions : • Connect all components. • Turn keyswitch ON. • Connect INSITE™ electronic service tool.		
Action	Specification/Repair	Next Step

Check for the appropriate circuit response after 30 seconds. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 381 inactive? YES Repair : None. The removal and installation of the connector corrected the fault.	4A
	Fault Code 381 inactive? NO Repair : Replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/82/82-019-031.html)	4A

STEP 4. Clear the fault code.

STEP 4A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Disable the fault code. - Start the engine and let it idle for 1 minute. - Use INSITE™ electronic service tool to verify that the fault code is inactive.	Fault Code 381 inactive? YES	4B
	Fault Code 381 inactive? NO Repair : Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	1A

STEP 4B. Clear the inactive fault codes.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Clear the inactive fault codes. Use INSITE™ electronic service tool to clear the inactive fault codes.	All fault codes cleared? YES	Repair complete
	All fault codes cleared? NO Repair : Troubleshoot any remaining active fault codes.	Appropriate troubleshooting steps